MIL-A-4637(USAF)
1 March 1953
Superseding
USAF No. 7538
22 September 1949

MILITARY SPECIFICATION

ATTENUATOR CN-29( )/UP, CN-42( )/UP, CN-43( )/UP

1. SCOPE

1.1 Type.- This specification covers three types fixed RF attenuators, designated Attenuator CN-29( )/UP, Attenuator CN-42( )/UP, and Attenuator CN-43( )/UP.

1.2 The parentheses in the nomenclature will be replaced on nameplates, et cetera, by a letter identifying the particular design; for example, Attenuator CN-29Z/UP. The identifying letter to be used to replace the parentheses will be furnished to the contractor upon his application to the Procuring Agency after the contract has been awarded.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-B-611	Brass, Commercial; Bars, Plates, Rods, Shapes, Sheets, and Strips
QQ-S-571	Solder, Soft (Tin, Tin-Lead, and Lead-Silver)

Military

MIL-C-71	Connectors, "N" For Radio Frequency Cables
MIL-P-116	Preservation, Methods of

U. S. Air Force

7200	Electronic Equipment; Aircraft, Preservation, Packaging, Packing and Marking For Shipment of

U. S. Army

72-53	Finishes (For Ground Signal Equipment)

STANDARD

Military

MIL-STD-129	Marking of Shipments

DRAWINGS

U. S. Air Force

D/L 53C54504	Attenuator, CN-29/UP, CN-42/UP, CN-43/UP, Assembly
50C13345	Plate, Identification

MWM CORP DAYTON, OHIO
5/60 300

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Signal Corps
SC-D-15914 Humidity Test Cycle For Ground Signal Equipment

Bureau of Ships
10-T-2145 Light Weight Shock Testing Machine

PUBLICATIONS
AMA Bulletin
No. 143 Specifications and Standards; Use of

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the Procuring Agency or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 General requirements.-

3.1.1 Specifications and standards.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143.

3.1.2 Finishes.- Finishes shall be in accordance with Specification No. 72-53.

3.1.3 Glass.- The glass inserts shall be thoroughly annealed and seasoned before the resistive coatings are applied and before the bullets are installed thereon.

3.1.4 Brass.- Brass shall be in accordance with Specification QQ-B-611.

3.1.5 Flux.- Rosin, rosin and alcohol, or equivalent plastic mixtures, shall be used as a flux. The flux shall be molten at a temperature below the solidus point of the solder. However, an acid flux may be used on mechanical assemblies provided that the parts are thoroughly and completely cleaned of all traces of the acid immediately after the soldering operation has been completed.

3.1.6 Solder and soldering.- All solder shall be in accordance with Specification QQ-S-571. All surfaces shall be cleaned before the application of solder. The metal surfaces while soldering shall attain a temperature above the solidus point of solder. Solder shall feather out to a thin edge, indicating proper flowing and wetting action. The solder shall not be crystallized, overheated or underheated. Soldering shall be neat and result in a good connection, both electrical and mechanical, at all joints. Soldering shall be performed so that no damage is caused to adjacent parts.

3.1.7 Mechanical assemblies.- No mechanical assembly shall depend solely on soft solder for mechanical strength or support.

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3.2 Service conditions.- The Attenuator CN-29( )/UP, Attenuator CN-42( )/UP and Attenuator CN-43( )/UP shall be so constructed that no fixed part or assembly shall become loose or be broken, and no degradation shall be caused in the performance below that specified by this specification when the attenuators are subjected to the following conditions:

3.2.1 Temperature.- Operation over the range of -54° to +71°C.

3.2.2 Humidity.- Operation at relative humidity as high as 100 percent at temperatures up to +66°C for both continuous and intermittent periods.

3.2.3 Vibration.- The attenuators shall be subjected to the vibration test in Section 4 of this specification.

3.2.4 Shock.- The attenuators shall be subjected to the shock test in Section 4 of this specification.

3.3 Life.- The attenuator shall be capable of continuous operation for a period of 4000 hours. The operational period may start after an initial storage period not exceeding two years.

3.4 Design.-

3.4.1 General.- The attenuators shall consist of a silver-plated brass casing approximately 5.1 inches long by 0.5 inch in diameter with a connector for a type "N" receptacle on one end (input) and a connector for a type "N" plug on the other end (output) and shall have a resistive center conducter permanently fixed in place.

3.4.2 Detail.- The detailed design of the attenuators shall be in accordance with the drawings of D/L 53C54504.

3.4.3 Precedence.- This specification, all drawings listed on D/L 53C54504, and the models of Attenuator CN-29/UP, Attenuator CN-42/UP, and Attenuator CN-43/UP, will be furnished to the contractor. In case of conflict, the order of precedence shall be: this specification, all drawings listed on D/L 53C54504, the models CN-29/UP, CN-42/UP, and CN-43/UP.

3.4.4 Nameplate.- The nameplate shall be in accordance with Drawing No. 50C13345. The information shall include the average power rating, frequency range, and the attenuation at three specified frequencies. The nameplate shall be curved and shall fit around the midsection of the attenuators.

3.4.5 Calibration.- The attenuators shall be calibrated for attenuation at 2.50, 2.70, and 3.00 kilomegacycles per second. The error of calibration shall not exceed 0.3 decibel at an ambient temperature of 25°C.

3.5 Performance requirements.-

3.5.1 Attenuation.-The attenuation of the Attenuator CN-43( )/UP shall be 16.0 ± 1.0 decibels at 25°C ambient temperature and shall attenuate a 2.70 kilomegacycle per second signal source.

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3.5.2 The attenuation of Attenuator CN-42( )/UP shall be 10.0 ± 1.0 decibels at 25°C ambient temperature and shall attenuate a 2.70 kilomegacycles per second signal source.

3.5.3 The attenuation of Attenuator CN-29( )/UP shall be 20.0 ± 1.0 decibels at 25°C ambient temperature and shall attenuate a 2.70 kilomegacycle per second signal source.

3.5.4 Power dissipation.- The attenuators shall be capable of continuous dissipation of up to and including one watt of average power over the entire specified frequency range. There shall be no voltage breakdown within the attenuator upon application of peak pulsed powers up to and including one kilowatt for pulse widths of 0.001 second during dissipation of one watt of average power, at any frequency within the specified frequency range.

3.5.5 Frequency sensitivity.- The allowable frequency sensitivity (variation in attenuation) shall be 1.4 decibels over the frequency range from 2.50 to 3.33 kilomegacycles for all attenuators.

3.5.6 Temperature sensitivity.- The attenuation of the attenuators shall not vary more than 0.14 decibel when subjected to a change of temperature (ambient) of 10°C.

3.5.7 Frequency range.- The attenuators shall be capable of operation over the range of 2.50 to 3.33 kilomegacycles per second.

3.5.8 Voltage standing wave ratio (VSWR).- The VSWR shall be less than 1.25:1 over the entire specified frequency and power ranges.

3.5.9 Nominal impedance.- The attenuator shall be matched to a nominal characteristic impedance of 49 ohms at both ends and shall be capable of being inserted in either a standard 5/16 inch rigid line of 49 ohms characteristic impedance or a standard cable of 50 ohms nominal characteristic impedance.

3.6 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.7 Workmanship.- All parts shall be manufactured and finished in a good workmanlike manner in accordance with the best commercial practice. All dimensions and tolerances shall be as specified in the detail drawings. All parts shall fit and operate in a manner that indicates good workmanship.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of attenuators shall be classified as follows:

(1) Qualification tests (see 4.2)
(2) Inspection tests (see 4.3)

4.2 Qualification tests.- Qualifications tests are those tests accomplished on samples submitted for qualification as a satisfactory prerequi-

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4.2.1 Prior qualification - Unless otherwise specified by the Procuring Agency, attenuators which have not previously passed a qualification test, or which have passed the qualification test and have been modified in any manner, shall satisfactorily pass a qualification test prior to the acceptance of any attenuators.

4.2.2 Qualification test samples.- Qualification test samples shall consist of the components listed below. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter.

(1) 1 each CN-29( )/UP Attenuator
(2) 1 each CN-42( )/UP Attenuator
(3) 1 each CN-43( )/UP Attenuator

4.2.3 The qualification test shall consist of the inspection tests and the following tests:

4.2.3.1 Extreme temperature test.- The attenuators shall be subjected to the following:

(1) Stabilize the attenuator temperature, and then measure performance at 21°C to 32°C.

(2) Reduce temperature to, and maintain in nonoperating condition for 24 hours at -54°C.

(3) Operate the attenuator at the temperature indicated in (2) for sufficient time to reach a stabilized condition and measure performance. Performance shall not be below the minimum requirements specified herein.

(4) Raise to room temperature and transfer to hot chamber.

(5) Raise temperature to, and maintain in nonoperating condition for 24 hours at 60°C.

(6) Operate the attenuator at the temperature indicated in (3) for a sufficient time to reach a stabilized condition and then measure performance. Performance shall not be below the minimum requirements specified herein.

(7) Raise to, and maintain in nonoperating condition for 4 hours at 71°C.

(8) Return to room temperature. Operate the attenuator for sufficient time to reach a stabilized condition and then measure performance. Performance shall be the same as in (1).

4.2.3.2 Humidity test.- The attenuators shall be subjected to the following humidity test:

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(1) Dry the attenuator at a temperature of 55° ± 3°C and a relative humidity of not greater than 30 percent for a period of 24 hours.

(2) Subject attenuator to a minimum of ten of the 48-hour cycles shown on Drawing SC-D-15914.

(3) Without removing from the humidity chamber, measure the performance during the periods indicated on Drawing SC-D-15914. Start measurements not more than five minutes after power has been applied. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(4) The performance in (3) shall not be below the minimum requirements specified herein.

4.2.3.3 Vibration test.-

4.2.3.3.1 Apparatus.- The vibration test shall be conducted on an apparatus capable of providing approximately sinusoidal vibration at all specified frequencies and amplitudes within 20 percent of specified values.

4.2.3.3.2 Variable frequency vibration.- The equipment shall be vibrated successively along each of the three mutually perpendicular directions that are respectively parallel to the edges of the equipment with an amplitude or acceleration and duration shown in the following tables, and a frequency which varies continuously between the limits shown in the table. The rate of change of the applied frequency shall not exceed one cycle per second. The attenuator shall be attached to the vibration table by a clamp around the center of the attenuator. No part of the attenuator shall touch any object other than the clamp. The tests may be interrupted at any time for inspection MCIP- electrical evaluation.

Method I	Freq. Limits	Amplitude (1/2 Total Excursion)	Duration in Each 3 Directions
	10-20 cps	1/32 inch	1 hour
	10-55 cps	1/64 inch	1 hour

Method II	Freq. Limits	Acceleration	Duration in Each 3 Directions
	10-20 cps	2 g	1/2 hour
	20-55 cps	3.5 g	1/2 hour

Either Method I (constant amplitude) or Method II (constant acceleration) may be used.

4.2.3.4 Shock test.-

4.2.3.4.1 Apparatus.- The shock test equipment shall be in accordance with Drawing No. 10-T-2145.

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4.2.3.4.2 Mounting.- The attenuators shall be secured rigidly to the platform of the shock test equipment by means of clamps around the midsection of the attenuators.

4.2.3.4.3 Test.- A total of nine blows shall be applied. Three blows shall be applied parallel to each of the three principal axes of the equipment being tested. The height of hammer drop from each back blow and each side blow shall be successively 1 foot, 2 feet and 3 feet. For each top blow, the height of the hammer drop shall be successively 2 feet, 3 feet and 4 feet. After each blow of the shock test, the attenuators shall continue to perform their functions without any attention to replacement of parts. The attenuators shall be capable of resuming normal operation after each blow. Minor damage or distortion shall be allowed, provided such damage or distortion shall not in any way impair the ability of the attenuators to perform their functions as specified herein.

4.3 Inspection tests.- Inspection of the attenuators shall consist of the following tests:

(1) Individual tests
(2) Sampling tests

4.3.1 Unless otherwise specified, contractor's records of all inspection work and test, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.3.2 The attenuators shall be subject to inspection by authorized Government Inspectors. When inspection is conducted at the contractor's plant, tests shall be conducted by the contractor, under the supervision of the authorized Government Inspector, and the report referred to herein shall be furnished to the Inspector.

4.3.3 Previous acceptance or approval of material or the release of any design by the Procuring Agency shall in no case be construed as a guarantee of the acceptance of the finished product.

4.3.4 Gauges and test equipment.- Gauges and test equipment which will be used by the contractor during inspection tests shall be approved by authorized representatives of the Procuring Agency prior to testing of the first unit from production. The gauges and test equipment shall be subject to inspection by authorized representatives of the Procuring Agency during production to determine that the gauges have not become worn to the extent that the fit of gauged parts will be adversely affected and that the test equipment retains the accuracy required for these tests.

4.3.4.1 Gauges.- Gauges shall be made of durable material, and shall be so dimensioned that gauged parts will be rejected if they are not within specified tolerances.

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4.3.4.2 Test equipment.- Test equipment to be used by the contractor in conducting the individual tests shall meet the following requirements:

4.3.4.2.1 Radio-frequency signal source.- The radio-frequency signal source shall consist of a tunable stable oscillator power supply and a square wave modulator, shall be capable of covering the frequency range of 2400 to 3700 megacycles per second and shall have an average power output of at least 30 to 50 milliwatts.

4.3.4.2.2 Amplifier.- The amplifier shall be a narrow band, tunable, high gain audio amplifier suitable for operation with a bolometer detector, and shall either include an internal output indicator or be used with an external output indicator.

4.3.4.2.3 Bolometer.- The bolometer and bolometer holder shall be matched for measurements over the frequency range of 2400 to 3700 megacycles per second.

4.3.4.2.4 Audio attenuator.- The audio attenuator shall be variable from 0 to 30 decibels, accurate to better than ±0.1 decibel at the modulator frequency and calibrated in tenths of a decibel.

4.3.4.2.5 Probe and slotted section.- The probe and slotted section shall be accurate to ± 2 percent with bolometer output.

4.3.4.2.6 Miscellaneous items.- Connecting cables, connectors and adapters as required shall be furnished.

4.3.5 Individual tests.- Each attenuator shall be subjected to the following tests:

4.3.5.1 Mechanical inspection.- The attenuators shall be given a thorough inspection for mechanical fit, uniformity of plating, poor workmanship, and other visual defects. The metallized glass tube insert shall be inspected by the contractor before assembly in the attenuator to determine that the resistive coating and varnish protecting coatings are properly applied, that the bullets are firmly and securely seated thereon, make proper contact with the resistive film, and that they have the resiliency and size required to insure good electrical contact when inserted in the jack-pin and plug-contact-pin of the attenuator casing.

4.3.5.2 Electrical test procedure.- The attenuators shall be electrically tested in accordance with procedures similar to those set forth below.

4.3.5.2.1 Calibration.-

(1) Connect the test equipment required by the paragraph headed "Gauges and test equipment" as shown by Detail B of Figure 1.

(2) Connect the radio-frequency cable from the radio-frequency signal source directly to the bolometer.

(3) Turn test equipment on and note the readings on the output indicating meter and on the audio attenuator.

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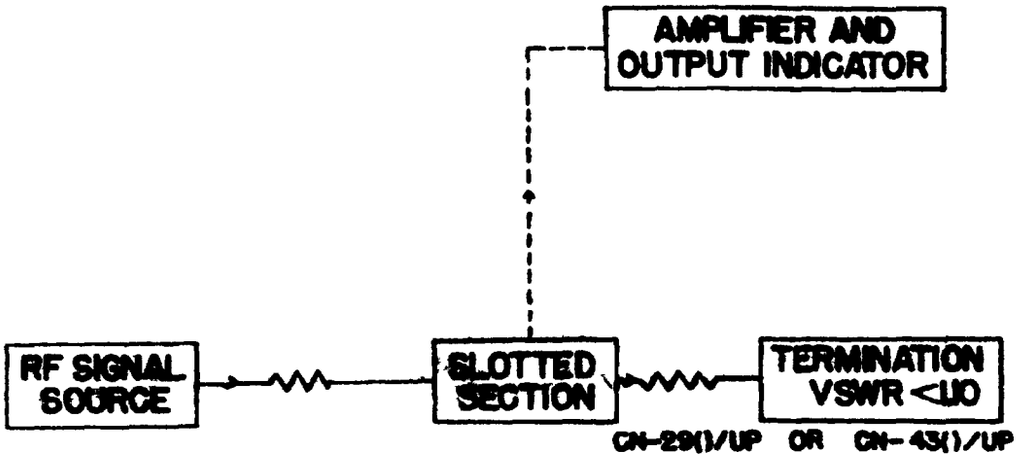


FIGURE 1 DETAIL A

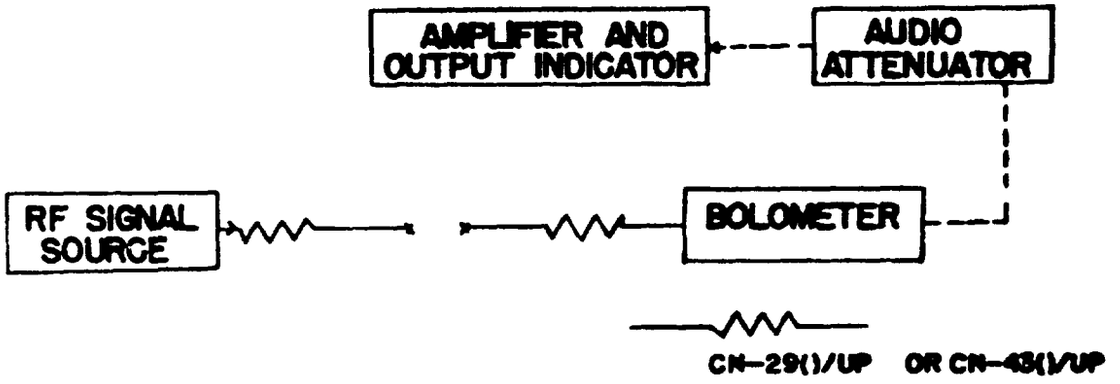


FIGURE 1 DETAIL B

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(4) Disconnect the oscillator radio-frequency cable from the bolometer.

(5) Connect the radio-frequency cable to the attenuator being tested and connect the attenuator to the bolometer.

(6) Adjust the audio attenuator until the output indicating meter shows the same reading as noted in (3) above.

(7) Read the attenuation as indicated by the change in dial reading of the audio attenuator.

(8) The reading shall conform to the requirements of 3.5.1.

4.3.5.2.2 Voltage standing wave ratio.-

(1) Connect the test equipment as shown in Detail A of Figure 1.

(2) Connect the output of the slotted section to the input connector of attenuator being tested.

(3) Turn test equipment on and read the voltage standing wave ratio on the output indicator.

(4) Disconnect the attenuator, reverse ends and repeat (2) and (3) above.

(5) The readings obtained in (2) and (3) above shall conform to the requirements of 3.5.8.

4.3.6 Sampling tests.- Sampling tests shall be performed as required by the Procuring Agency. A complete description of the procedure proposed for conducting sampling tests shall be submitted to the Procuring Agency for approval. The proposed testing procedure shall determine compliance with the requirements of 3.5.4, 3.5.5, and 3.5.6.

4.4 Rejections.-

4.4.1 The individual attenuators failing to meet their respective tests shall be rejected.

4.4.2 When tests are specified on a quantity of attenuators that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional attenuators of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the attenuators. When the cause of failure has been determined, the contractor shall incorporate the changes necessary to correct the defects in the remaining attenuators on the order.

4.4.3 Rejected attenuators either shall be replaced with other attenuators or shall be reworked to correct the defects, after which all specified tests shall be repeated.

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4.4.4 Rejected attenuators shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

5. PREPARATION FOR DELIVERY

5.1 General.- The packaging, packing and marking requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 The attenuators shall be packaged and packed for shipment in accordance with the applicable parts of Specification No. 7200 except that Specification MIL-P-116 shall replace Specification AN-P-13.

5.3 Marking.- Interior and exterior markings shall be in accordance with MIL-STD-129. The nomenclature shall be Attenuator CN-29( )/UP, CN-42( )/UP, or CN-43( )/UP as applicable, mfrs part no., Specification MIL-A-4637, To be supplied by the contractor.

6. NOTES

6.1 Use.-

6.1.1 Attenuator CN-43( )/UP is a fixed RF attenuator for a 5/16 inch coaxial line with Type "N" connectors that will be used to decrease the power of RF signals in the frequency range from 2.40 to 3.70 kilomegacycles per second by a ratio of 40 to 1 (16 decibels).

6.1.2 Attenuator CN-29( )/UP is a fixed RF attenuator for a 5/16 inch coaxial line with Type "N" connectors that will be used to decrease the power of RF signals in the frequency range from 2.40 to 3.70 kilomegacycles per second by a ratio of 100 to 1 (20 decibels).

6.1.3 Attenuator CN-42( )/UP is a fixed RF attenuator for a 5/16 inch coaxial line with Type "N" connectors that will be used to decrease the power of RF signals in the frequency range from 2.40 to 3.70 kilomegacycles per second by a ratio of 10 to 1 (10 decibels).

6.2 Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of supplier is called to this requirement, and manufacturers are urged to communicate with the Rome Air Development Center and arrange to have the products that they propose to offer to the Air Force tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commanding General, Rome Air Development Center, Griffiss Air Force Base, Rome, New York, attention: RCRES2D.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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